

23 Lecture 23: Factorization¹

23.1 Two theorems in number theory²

Theorem 1: if N is a composite (i.e. not prime) L -bit integer, and y is a solution of $y^2 \equiv 1 \pmod{N}$ in the interval $1 < y < N - 1$ (thus the trivial solutions $y = 1, y = N - 1 \equiv -1 \pmod{N}$ are excluded), then $\text{GCD}(y - 1, N)$ and $\text{GCD}(y + 1, N)$ are nontrivial factors of N that can be computed with $O(L^3)$ operations.³

Proof: since $y^2 \equiv 1 \pmod{N}$, N must divide $y^2 - 1 = (y + 1)(y - 1)$. Moreover $N > y + 1 > y - 1$ and therefore N cannot divide $(y + 1)$ or $(y - 1)$, but must have nontrivial common factors with $(y + 1)$ and $(y - 1)$. These common factors can be found by computing the GCD of N and $y \pm 1$ with Euclid's algorithm, which requires $O(L^3)$ operations.⁴

Theorem 2: if we choose at random an integer x in the interval $[1, N - 1]$, coprime with N , it is very probable that its order $(\text{mod } N)$ is even, and $x^{r/2}$ is *not* equal to $\pm 1 \pmod{N}$ (the probability is always $\geq 1/2$, see Appendix 4 of Nielsen and Chuang, where this probability is shown to be $1 - \frac{1}{2^m}$ where m is the number of different prime factors of N).⁵

Then $y = x^{r/2} \pmod{N}$ is a nontrivial solution of $y^2 \equiv 1 \pmod{N}$.⁶

In conclusion, finding the order r modulo N of an integer x , chosen at random in the interval $[1, N - 1]$, and coprime with N , allows to find in polynomial time a nontrivial solution of $y^2 \equiv 1 \pmod{N}$, and hence a nontrivial factor of N .⁷

23.2 The factorization algorithm⁸

We have seen that the two theorems of the previous section can be combined to construct a factorization algorithm. This algorithm was first proposed by P.W. Shor in 1994, and produces in polynomial time and with good probability a nontrivial factor of a composite integer N . All operations can be executed on a classical computer, except the order finding subroutine that requires a quantum computer (to find the order in polynomial time).⁹

23.3 Summary of the algorithm¹⁰

- **Input:** N ¹¹
- **Output:** a nontrivial factor of N
- **Runtime:** $O((\log_2 N)^3)$ operations

- **Procedure:** ¹²

1. If N even. $\rightarrow$ output = 2 1
2. Determine whether $N = a^b$ for $a \geq 1, b \geq 2$ (can be done by a classical algorithm in $O(L^3)$ operations). If yes, $\rightarrow$ output = a .
3. Choose x in the interval $[1, N - 1]$. Compute the $\text{GCD}(x, N)$, if it is > 1 then $\rightarrow$ output = $\text{GCD}(x, N)$
4. Quantum subroutine to find the order r of x modulo N .
5. If r is even and $x^{r/2} \not\equiv -1 \pmod{N}$, compute $\text{GCD}(x^{r/2} - 1, N)$ and $\text{GCD}(x^{r/2} + 1, N)$, and test whether these are nontrivial factors of N . If yes, $\rightarrow$ output = these factors. 2
- NB:** $x^{r/2}$ cannot be $\equiv +1 \pmod{N}$ because r is the *smallest* integer such that $x^r \equiv 1 \pmod{N}$. 3

23.4 Example: the factorization of $N = 91$ 4

- i) The steps 1. and 2. are passed without exiting. 5
- ii) step 3. : choose for example $x = 4$, coprime with 91.
- iii) compute order of 4 modulo 91. Result = 6 ($4^6 = 4096 = 1 + 45 \times 91$).
- iv) check that $x^{r/2} \not\equiv -1 \pmod{N}$, $\rightarrow 4^3 = 64 \not\equiv -1 \pmod{91}$, ok.
- v) $\text{GCD}(64+1, 91) = 13$, $\text{GCD}(64-1, 91) = 7$

23.5 Bibliography 6

P. W. Shor (1994), "Algorithms for quantum computation: discrete log and factoring", Proc. of the 35-th Annual Symposium on the Foundations of Computer Science, 124-134, IEEE Computer Society Press. 7